Problem sheet 3

1. A random variable X has pdf f(x), where

$$f(x) = \begin{cases} x/8 & 0 \le x \le 2 \\ 1/4 & 2 \le x \le 4 \\ 3/4 - x/8 & 4 \le x \le 6 \end{cases}$$

Show that this is a valid p.d.f, and find $E(X)$ and $Var(X)$ (ans: 3, 5/3)

$$F(\infty) = P(X \le \infty) = \int_{-\infty}^{\infty} f(x)\,dx$$

$$= \int_0^2 \frac{1}{8}x\,dx + \int_2^4 \frac{1}{4}\,dx + \int_4^6 \left(\frac{3}{4} - \frac{1}{8}x\right)dx$$

$$= \left[\frac{1}{16}x^2\right]_0^2 + \left[\frac{1}{4}x\right]_2^4 + \left[\frac{3}{4}x - \frac{1}{16}x^2\right]_4^6$$

$$= \left(\frac{1}{4} - 0\right) + \left(1 - \frac{1}{2}\right) + \left(\frac{9}{2} - \frac{9}{4} - 3 + 1\right)$$

$$= \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$$ and $f(x) \ge 0$

$\therefore$ valid p.d.f.

$$E(X) = \int_{-\infty}^{\infty} x f(x)\,dx =$$

$$= \int_0^2 \frac{1}{8}x^2\,dx + \int_2^4 \frac{1}{4}x\,dx + \int_4^6 \left(\frac{3}{4}x - \frac{1}{8}x^2\right)dx$$

$$= \frac{1}{3} + \frac{3}{2} + \frac{7}{6} = 3$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx$$

$$= \int_0^2 \frac{1}{8}x^3\,dx + \int_2^4 \frac{1}{4}x^2\,dx + \int_4^6 \left(\frac{3}{4}x^2 - \frac{1}{8}x^3\right)dx$$

$$= \frac{1}{2} + \frac{14}{3} + \frac{11}{2} = \frac{32}{3}$$

$$\therefore Var(X) = E(X^2) - [E(X)]^2$$

$$= \frac{5}{3}$$

2. The absolute error X (in cm) of a measurement from an instrument has p.d.f given by

$$f(x) = \frac{2}{\alpha}\left(1 - \frac{x}{\alpha}\right)$$

where α is the upper limit of the absolute error.

Show that $f(x)$ is a valid p.d.f and assuming $\alpha = 0.005$, find $E(X)$ and $Var(X)$.

(0.00167, 0.0000013)

$$\int_{-\infty}^{\infty} f(x)\,dx = \left[\frac{2}{\alpha}x - \frac{1}{\alpha^2}x^2\right]_0^{\alpha}$$

$$= 2 - 1 = 1$$

$$E(X) = \int_{-\infty}^{\infty} x f(x)\,dx = \left[\frac{1}{\alpha}x^2 - \frac{2}{3\alpha^2}x^3\right]_0^{\alpha}$$

$$= \alpha - \frac{2}{3}\alpha = \frac{1}{3}\alpha = 0.001\dot{6}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \left[\frac{2}{3\alpha}x^3 - \frac{1}{2\alpha^2}x^4\right]_0^{\alpha}$$

$$= \frac{2}{3}\alpha^2 - \frac{1}{2}\alpha^2 = \frac{1}{6}\alpha^2 = 4.1\dot{6} \times 10^{-6}$$

$$\therefore Var(X) = E(X^2) - [E(X)]^2 = 1.3\dot{8} \times 10^{-6}$$

3. A pilot, A, has to pass a particular test to qualify. If A fails the test on the first attempt, further (independent) tests are allowed until A passes. Show that X, the total number of attempts required, has probability distribution

$$P(X = x) = (1-p)^{x-1}p, \quad x = 1,2,3 \ldots$$

where p is the probability of passing each individual test.

Show that $E(X) = 1 - p$.

$$P(X = x) = p \cdot \prod_{k=1}^{x-1} (\bar{p})$$

$$= (1-p)^{x-1} p$$

$$E(X) = \sum_{i=1}^{N} x_i P(X = x_i)$$

$$= \sum_{i=1}^{N} x_i (1-p)^{x_i - 1} p$$

$$= p \sum_{i=1}^{N} x_i (1-p)^{x_i - 1}$$

$$= p \sum_{i=1}^{N} -\frac{d}{dp}[(1-p)^{x_i}]$$

$$= -p \frac{d}{dp}\left[\sum_{i=1}^{N} (1-p)^{x_i}\right]$$

$$= -p \cdot \frac{d}{dp}\left[\frac{1-p}{1-(1-p)}\right]$$

$$= -p \cdot \frac{d}{dp}\left[\frac{1-p}{p}\right]$$

$$= -p\left(\frac{-1}{p^2}\right) = \frac{1}{p}$$

4. A company manufactures batches of 1000 similar but independent aircraft components. The probability of a fault in any component is 0.004.
Find an approximate value for the probability that
(a) two different batches both contain more than 3 defective items, (0.479)
(b) 50 batches between them contain fewer than 15 defective items.
(show they are rubbish)

$n = 1000,\ p = 0.004$

$\therefore\ np = 4$

Using normal distribution to approximate binomial distribution, so $N(4, 1.996)$

(a) $P(X \geq 3) = 1 - \Phi\left(x \leq \frac{3-4}{1.996}\right)$

$= 1 - \Phi(x \leq -0.501)$

$= 1 - [1 - \Phi(x \geq 0.501)]$

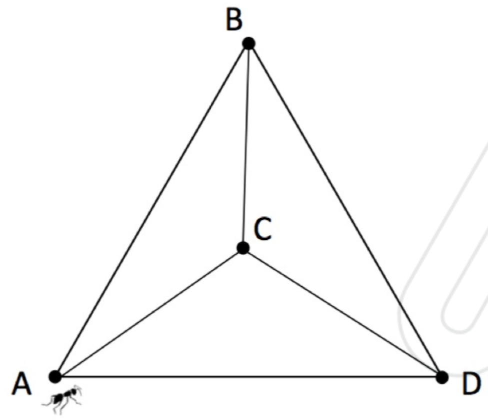
≈ 0.6915

$\therefore\ P(X \geq 3) \cap P(X \geq 3) = (0.6915)^2 = 0.478$

(b) $P(X \leq 15) = \Phi(X \leq 5.51) \approx 1$

$\therefore\ \prod_{k=1}^{50} P(X \leq 15) \approx 1^{50} = 1$

5. An ant moves at random along the edges the figure $ABCD$. It starts at the vertex A, then moves with equal probability to one of the other verticies (i.e., it has probability $\frac{1}{3}$ of moving to vertex B, C or D). The ant stops if it reaches the centre C and continues moving otherwise.



Let S be the total number of moves taken by the ant.

a. Show that $P(S = n) = \frac{2^{n-1}}{3^n}$

b. Find the expected number of moves $E(S)$. (Ans 3)

[Hint: for $0 < x < 1$, it is the case that $\sum_{n=1}^{\infty} nx^n = \frac{x}{(1-x)^2}$]

(a) $p = \frac{1}{3}$

$\because P(S=1) = \frac{1}{3}$

$P(S=2) = 2 \cdot \frac{1}{3} \cdot P(S=1) = 2 \cdot (\frac{1}{3})^2$

$P(S=3) = 2 \cdot \frac{1}{3} \cdot P(S=2) = 2^2 \cdot (\frac{1}{3})^3$

...

$\therefore P(S=n) = \frac{2^{n-1}}{3^n}$

(b) $E(S) = \sum_{i=1}^{N} n_i P(S=n_i) = \sum_{i=1}^{N} n_i \frac{2^{n_i-1}}{3^{n_i}}$

$= \sum_{i=1}^{N} \frac{1}{2} n_i (\frac{2}{3})^{n_i} = \frac{1}{2} \cdot \frac{\frac{2}{3}}{(1-\frac{2}{3})^2} = 3$

6. The price of a stock is modelled to rise 1p per day with probability $\frac{3}{4}$ and fall 1p per day with probability $\frac{1}{4}$. Let X_i be a random variable with distribution

$$P(X_i = 1) = \frac{3}{4} \quad \text{and} \quad P(X_i = -1) = \frac{1}{4}$$

The current price of the stock is £1 and price movements between days are assumed to be independent.

a. Explain why the price in pence after n days is modelled by the random variable

$$Y_n = 100 + \sum_{i=1}^{n} X_i$$

b. Find $E(X_i)$ and $Var(X_i)$ Ans (1/2, ¾)

c. Use the central limit theorem to approximate Y_n by a normal distribution for large n.

d. Use your result in (c) to approximate the probability that the stock price will be greater than £2.55 after 300 days Ans (0.3707)

a. Change in price after n-day is $\sum_{i=1}^{n} X_i$

Initial price is 1 (£) = 100 (p)

$\therefore Y_n = 100 + \sum_{i=1}^{n} X_i$ (p)

b. $E(X) = \sum_{i=1}^{n} x_i P(X = x_i) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$

$E(X^2) = \frac{3}{4} + \frac{1}{4} = 1$

$\therefore Var(X) = E(X^2) - [E(X)]^2 = \frac{3}{4}$

c. for large n,

100 +

$$Y_n \approx 100 + N\left(\frac{1}{2}n, \frac{3}{4}n\right)$$

d. for $n = 300$,

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$$Y_n \approx 100 + N(150, 225)$$

X   X   255

$$\therefore P(Y_n > 255) = 1 - P(N \le 155)$$

255   250

$$= 1 - \Phi\left(X \le \frac{155 - 150}{15}\right)$$

$$= 1 - \Phi\left(X \le \frac{1}{3}\right)$$

$$\approx 1 - \Phi(X \le 0.33)$$

$$= 1 - 0.6293$$

$$= 0.3707$$

7. An aircraft component has a lifetime, which is exponentially distributed with mean 2000 hours. Find the probability that it lasts more than 1000 hours. Two independent components of this type are arranged in series to form a device that will operate if and only if both components function.
   a. What is the probability that the device will operate after 1000 hours of use? (Ans 0.606)
   b. Find the hazard rate (ans 1/1000) and hazard function of the device.
   c. What is the expected lifetime of the device? (1000 hrs)

a. $E(X) = \frac{1}{\lambda} = 2000 \ (h)$

$\therefore \lambda = \frac{1}{2000}$

$$\therefore f(t) = \begin{cases} \frac{1}{2000} e^{-\frac{t}{2000}} & 0 \le t \le \infty \\ 0 & \text{otherwise} \end{cases}$$

$\therefore F(t) = \int_0^t f(t)\,dt = 1 - e^{-\frac{t}{2000}}$

$\therefore S(t) = 1 - F(t) = e^{-\frac{t}{2000}}$ $\quad \therefore S(1000) = 0.6065$

$\therefore S_{total}(t) = S(t) \cdot S(t) = e^{-\frac{t}{1000}}$

b. $H(t) = -\ln[S_{total}(t)] = \frac{t}{1000}$

$\therefore h(t) = \frac{dH(t)}{dt} = \frac{1}{1000}$

c. $E(T) = \int_0^\infty t \cdot f(t)\,dt = -\int_0^\infty t \cdot S'(t)\,dt$

$= [t \cdot S(t)]_0^\infty - \int_0^\infty S(t)\,dt = 1000 \ (h)$

8. Suppose that the components in question 7 are now arranged in parallel to form a new device that will operate if and only if at least one of the two components functions.
    a. What is the probability that this new device will operate after 1000 hours of use? (0.845)
    b. Find the hazard rate and hazard function of the new device.
    c. What is the expected lifetime of the new device? (3000 hrs)

a. $S_{total}(t) = 1 - [1 - S(t)][1 - S(t)]$

$$= 1 - (1 - 2e^{-\frac{t}{2000}} + e^{-\frac{t}{1000}})$$

$$= 2e^{-\frac{t}{2000}} - e^{-\frac{t}{1000}}$$

$\therefore S_{total}(1000) = 0.8452$

b. $H(t) = -\ln[S_{total}(t)]$

$$= -\ln[2e^{-\frac{t}{2000}} - e^{-\frac{t}{1000}}]$$

$$= -\ln 2 \ln(e^{-\frac{t}{2000}})[-\ln(e^{-\frac{t}{1000}})]$$

$$= \frac{t^2}{2000000}\ln 2$$

$$h(t) = \frac{dH}{dt} = \frac{t}{1000000}\ln 2$$

c. $E(T) = \int_0^\infty t f(t)\,dt = -\int_0^\infty t S'(t)\,dt$

$$= [tS(t)]_0^\infty - \int_0^\infty S(t)\,dt = 3000(h)$$